



NATIONAL INSTITUTE OF TECHNOLOGY ROURKELA-769008

MID-SEMESTER EXAMINATION, 2023

SESSION: 2022 – 2023 (Spring)

B. Tech. 8th / M. Tech. 2nd Semester

Subject code: CS6404

No. of pages: 2

Subject Name: Software Testing

Full Marks: 60

Dept. Code: CS

Duration: 2hrs

**Figures at the right hand margin indicate marks.
All parts of a question should be answered at one place.
Answer all questions.**

Q. No.	Particulars	Marks
1. (a)	Suppose 10 different test case design strategies are successively used to test a program. Each test case design strategy is capable of detecting 40% of the bugs existing at the time of application of the strategy. If 1000 bugs existed in the program, determine the number of bugs existing at the end of testing.	2*10=20
(b)	At least how many test cases are required to achieve MC/DC coverage of the following code segment if((a>5 and (b<100) and (c>50)) x = x+1;	
(c)	Give a positive and a negative test scenario for the “Date of Birth” field of the Student Enrollment Page in NIT Rourkela website.	
(d)	A defect –seeding program inserts 81 defects into an application. Inspections and testing found 152 defects. Of these, 72 were seeded defects. What is the expected number of defects that remain undetected in the application?	
(e)	What do you mean by equivalent mutants? Give an example of equivalent mutants.	
(f)	Give an example where it is not possible to determine the state of data variable by just static analysis of the code.	
(g)	Suppose a program contains 5 decision inputs, if there are 4 choices at each decision point, how many test cases are necessary for branch testing?	
(h)	What do you mean by concolic testing? Give an example of concolic tester.	
(i)	Give an example of a bug that is detected by the black-box test suite, but is not detected by the white-box test suite. Justify your answer.	
(j)	Suppose two programmers are assigned the same programming problem and they develop this independently. Explain how you can compare their programs with respect to understanding difficulty.	
2. (a)	A mobile phone service provider uses a program that computes the monthly bill of customers as follows: a. Minimum Rs 400 for up to 150 calls b. Plus Rs 1 per call for the next 80 calls c. Plus Rs 0.50 per call for the next 70 calls d. Plus Rs 0.30 per call for any call beyond 300 calls Design test cases for this program using equivalence class testing technique.	10+5+5=20
(b)	A program calculates the GCD of three numbers in the range [1, 50]. Design test cases for this program using robust testing, and worst-case testing methods. Consider the following system:	

(c)	Passengers who travel more than 50,000 km per calendar year in addition, pay cash for tickets or have been travelling regularly for more than 8 years are to receive free round trip ticket around India. Passengers who travel less than 50,000 km per calendar year and have been availing railway service regularly for more than 8 years also get a free round ticket around India. Design test cases for this system using decision table based testing.	
3.	Consider the following program	10+10=20
(a)	<pre>#include <stdio.h> main{ int a,b,c,d; clrscr(); printf("Enter the two variables a, b"); scanf("%d %d", &a , &b); printf("Enter the option 1:Addition, 2:Subtraction, 3:Multiplication, 4:Division"); scanf("%d",&c); switch(c){ case1:d= a+b; printf("Addition of the two no.=%d",d); break; case2:d= a-b; printf("Subtraction of the two no.=%d",d); break; case3:d= a*b; printf("Multiplication of the two no.=%d",d); break; case4:d= a/b; printf("Division of the two no.=%d",d); break; } }</pre> - Draw the CFG for the above program. - Calculate the cyclomatic complexity of the program. - List all the independent paths. - Design all the test cases from the independent paths.	
(b)	Consider the program given in part (a) and perform static data flow analysis by finding out the <i>define-use-kill</i> patterns for all the variables in the source code.	